HARSH DUBEY

AI Full Stack Developer

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PROFESSIONAL SUMMARY

AI Full Stack Developer with production experience building LLM powered product features across React, Next.js, and FastAPI, from RAG ingestion and API layer through to streaming, responsive user interfaces. Comfortable owning a feature end to end.

PROFESSIONAL EXPERIENCE

BNCW ENTERPRISES | Frontend Developer **September 2025 to September 2026**

Full stack contributor on a Next.js and FastAPI platform serving an LLM powered product, owning frontend surfaces while shipping backend endpoints, integrations, and RAG pipeline work.

- **AI Chat Feature, End to End:** Built an **LLM powered** research assistant from the FastAPI retrieval endpoint through to the **React** interface, handling multi turn conversation state, request lifecycle, and error recovery.
- **RAG Ingestion Pipeline:** Extended the document ingestion pipeline for a **pgvector** backed **RAG** service in Python, covering source contracts, content preview, manual sync, and backfill paths that keep the retrieval corpus current.
- **Streaming and Safe Rendering:** Implemented the streaming response layer over **Server Sent Events**, rendering model generated markdown through a sanitized pipeline that blocks script injection from untrusted output.
- **Backend for Frontend Layer:** Built **Next.js** API routes as a backend for frontend layer over external market data providers and internal services, keeping provider credentials server side and exposing one typed contract to the client.
- **REST API Development:** Shipped **FastAPI** endpoints with **SQLAlchemy** models and **Pydantic** schemas, covering route design, permission guards, and consistent error contracts, with **pytest** coverage on new paths.
- **Third Party Integrations:** Integrated **AWS S3** for media upload, **AWS SES** for transactional email, and external market data providers, wiring webhook and API flows between backend services and the client.
- **Frontend Engineering:** Built reusable component libraries and responsive layouts with **Redux Toolkit**, **Tailwind CSS**, and **Material UI**, delivering UI surfaces including a rich text editor, drag-and-drop reordering, and interactive **Recharts** dashboards unit-tested with **Vitest**.
- **State and Authentication:** Structured the Redux Toolkit store and client data layer, and built login, registration, password reset, and email verification flows against **JWT** backed APIs with route guards and session handling.

PROJECTS

RouteLens | TypeScript, Node.js, VS Code Extension API, MCP, Zod, Vitest, GitHub Actions

- Built an open source **VS Code extension** that finds every API route in **Next.js** and **Express** codebases through static analysis with the **TypeScript compiler API**, and generates sample requests that pass each route's **Zod** schema.
- Shipped one click testing of every route with re-runs on save, multi step request flows, and automatic login backed by the OS keychain, plus a **CLI** that runs the same suites in **GitHub Actions** with JUnit reports.
- Built an **MCP server** with 7 tools so AI agents like **Claude Code** and **GitHub Copilot** can run the API, read each failure with its handler source, and fix it behind host guard rails, covered by about 380 tests.

AI Powered Mock Interview Platform | React 19, TypeScript, Firebase Auth, Firestore, Google Gemini AI, Web Speech API

- Built an AI mock interview platform that generates role specific questions grounded in the candidate's uploaded **CV**, captures spoken answers through the webcam with live **speech to text**, and scores each answer against a model answer with the specific gap named.
- Integrated **Google Gemini** with **structured output** schemas and native PDF input, behind a model fallback chain that treats quota (429) and overload (503) errors differently to stretch free tier limits.
- Implemented **Firebase Authentication** with email and Google sign in, and per user **Firestore security rules** keyed on the auth UID, so access control needs no separate backend or token exchange.

EDUCATION

United College of Engineering and Research, Prayagraj | Bachelor of Technology in Computer Science **2021 to 2025**

TECHNICAL SKILLS

AI and LLM: RAG Pipelines, Document Ingestion, Vector Search, pgvector, Embeddings, Prompt Engineering, LLM Integration, MCP Servers, Anthropic Claude, OpenAI, Google Gemini, Langfuse

Backend: Python, FastAPI, Node.js, Express.js, RESTful APIs, SQLAlchemy, Pydantic, JWT Authentication, Webhooks, pytest

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), Redux Toolkit, Tailwind CSS, Material UI, Shadcn UI, HTML5, CSS3, Responsive Design, Vitest, React Testing Library

Databases: PostgreSQL, pgvector, MySQL, MongoDB, Firestore

Cloud and Tools: AWS (S3, SES, RDS), Cloudflare Workers, Docker, Git, GitHub Actions, Bitbucket, Stripe, Sentry, PostHog, Jira